Saket Sandeep Deshmukh

AI / GenAI Engineer

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SUMMARY

AI Engineer with **2.5+ years of experience** building and shipping production-grade LLM, RAG, and Agentic AI systems in regulated domains (fintech and healthcare). Worked directly with stakeholders to translate business requirements and AI PoCs into scalable, production-ready solutions.

SKILLS

Programming: Python, SQL, Bash

Generative AI & LLMs: LLMs, RAG, Agentic AI, Prompt Engineering, Structured Outputs, Function Calling, Model Routing, Grounding, Guardrails, LLM Evaluation (TruLens)

AI Frameworks & APIs: LangChain, LangGraph, LiteLLM, FastAPI, Pydantic, MCP, OpenAI API

AI Platforms & Models: Amazon Bedrock, Amazon Bedrock Guardrails, vLLM, Ollama, DeepSeek API

Document AI: Amazon Textract, OCR, PyMuPDF, Vision Models, PII Masking

Retrieval & Databases: PostgreSQL, pgvector, FAISS, BM25, Hybrid Search, Embeddings, MySQL

Cloud & DevOps: AWS, Docker, Kubernetes, Git, GitHub Actions, Jenkins, Ansible

EXPERIENCE

KWAD Corporation Pvt. Ltd. – Software Engineer **Apr 2025 – Present**

Fintech – AI Compliance & Financial Operations Copilot

- Built a production **agentic RAG** system that helps operations and compliance teams investigate B2B transactions across 12 tenants using natural-language queries.
- Developed **LangGraph** workflows for intent/risk classification, conditional routing, tool calling, retries, and human escalation.
- Designed document processing with PyMuPDF, Amazon Textract OCR, quality checks, and vision-model fallback, routing unreadable pages to human review.
- Implemented **SQL-first retrieval** with metadata filtering, tenant-level access restrictions, and hybrid search to improve the relevance and accuracy of retrieved evidence.
- Connected PostgreSQL/pgvector retrieval, transaction SQL tools, compliance evidence, and deterministic rules with GPT-4o and LiteLLM.
- Implemented FastAPI authentication, RBAC, and tenant-aware access controls, with source citations and human review for high-risk cases.
- Improved evaluation and observability through test sets, tracing, dashboards, and regression checks; 9 optimization iterations reduced hallucinated responses by **38%** and regression detection time by **45%**.
- Automated tests, security checks, Docker builds, and **Amazon EKS** deployments with GitHub Actions CI/CD.

Healthcare – AI Platform for Clinical Record Processing

- Collaborated with stakeholders to define extraction schemas and clinical validation rules, and built a **Python/FastAPI** platform on AWS using **Amazon S3** to process **5K+** scanned, handwritten, and multilingual clinical records into structured data with human-review workflows.
- Integrated **Amazon Textract** for OCR and **Amazon Bedrock** for LLM-based extraction, model routing, and structured outputs, achieving **99% schema compliance** through Pydantic and JSON Schema validation.
- Developed a deterministic grounding and clinical validation layer to verify extracted values against source documents, automatically validating **75% of fields** and routing exceptions to human reviewers with source references.
- Implemented AI safety controls for **prompt-injection detection, PII masking, citation verification, and confidence-based routing** to improve extraction reliability and protect patient data.
- Designed document-processing workflows with asynchronous message handling and failure recovery using **Amazon SQS**, supporting reliable processing of large document batches.

Briowin Private Limited – Software Engineer **Feb 2024 – Mar 2025**

E-Commerce – AI-Powered Product Information Assistant

- Built an AI product assistant for a 300+ SKU catalog, reducing query resolution time by **52%**.
- Developed a LangGraph workflow with validation, fallback handling, and AI-based optimizations, improving response relevance by **30%** and achieving a **94%** request success rate.
- Built a hybrid retrieval system combining OpenAI `text-embedding-3-small` semantic search and BM25 keyword search in parallel, improving recall by **22%**, precision by **28%**, and reducing latency by **33%**.
- Integrated **MCP** with catalog services and Reciprocal Rank Fusion to improve top-k retrieval accuracy by **27%**.
- Added input/output guardrails to filter irrelevant queries and reduce hallucinations by **35%**, while cutting unnecessary LLM and embedding calls.
- Built an automated **RAGAS** evaluation pipeline to monitor context precision, context recall, relevance, faithfulness, and groundedness.
- Maintained Jenkins CI/CD, automated deployments with Ansible, and deployed applications on Kubernetes.

EDUCATION

Bachelors in Computer Engineering – Savitribai Phule Pune University **2019 – 2023**

CERTIFICATIONS

- Evaluating RAG and Semantic Search Systems – NVIDIA
- OCI AI Foundations Associate – Oracle
- Linux Red Hat Certified Engineer (RHCE - EX294) – Udemy
- LangChain Mastery – Udemy

AWARDS

- **Product and Platforms Pinnacle Award** – KWAD Corporation, 2026